1.3 limit properties

1.3.1

$\lim_{x\to c} f(x) = \boxed{L}$ and $\lim_{x\to c} g(x) = \boxed{K}$

$\frac{f(x)}{g(x)} = \frac{L}{K}$ $\quad \pm \to \pm$

$f(x) \cdot g(x) = LK$

Constants: $\lim_{x\to c} \boxed{b = b}$

identity: $\lim_{x\to c} \boxed{x = c}$

sum/difference: $\lim_{x\to c} (f(x) \pm g(x)) = L \pm K$

scalar multiples: $\lim_{x\to c} \boxed{b \cdot f(x)} = bL$

products: $\lim_{x\to c} \boxed{f(x) \cdot g(x)} = LK$

Quotients: $\lim_{x\to c} \boxed{\frac{f(x)}{g(x)}} = \boxed{\frac{L}{K}}$ $(K \neq 0)$

powers: $\lim_{x\to c} \boxed{f(x)}^n = L^n$

$f(x) \to$ sub for L

Roots: $\lim_{x\to c} \sqrt[n]{f(x)} = \sqrt[n]{L}$

*Composition: $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = K$, $g(L) = K$

then $\lim_{x\to c} g(f(x)) = K$

$g(x) = k$

$g(L) = k$

$g(f(x)) = k$

1.3.2

1. $\lim_{x\to c} p(x) = p(c)$

2. $\lim_{x\to c} \frac{p(x)}{q(x)} = \frac{p(c)}{q(c)} \to \neq 0$

1.3.3 · special limits

1. $\lim_{x\to c} \sin x = \sin c$
2. $\lim_{x\to c} \cos x = \cos c$
3. $\lim_{x\to c} \tan x = \tan c$
4. $\lim_{x\to c} \csc x = \csc c$
5. $\lim_{x\to c} \sec x = \sec c$
6. $\lim_{x\to c} \cot x = \cot c$

7.

8.

9.

sub $\boxed{x}$ with $\boxed{c}$

. $\lim_{x\to c} a^x = a^c$

. $\lim_{x\to c} \ln x = \ln c$

$\lim_{x\to c} \sqrt[n]{x} = \sqrt[n]{c}$

1.3.4 Squeeze theorem

$\underbrace{\boxed{f(x)} \le g(x) \le \boxed{h(x)}}_{} \; \overset{=L}{\longrightarrow} \lim_{x\to c} g(x) = L$

if $\lim_{x\to c} f(x) = L = \lim_{x\to c} h(x)$

then $\lim_{x\to c} g(x) = L$

1.3.5 Special limits

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1$
2. $\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$
3. $\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$
4. $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$

1.3.6 → limits of function at all but one point

$g(x) = f(x)$ for x in open interval except possibly at $\boxed{c}$

$\lim_{x\to c} g(x) = L$ (L is real number)

↳ $\lim_{x\to c} f(x) = L$

1.3.7 ✓

↳ Another way to find "$\frac{0}{0}$"

difference in quotients